

9(a)(i)	emission of electrons (from a metal surface)	B1
	when electromagnetic radiation is incident (on electrons)	B1
9a(ii)	<u>minimum</u> energy required for an electron to leave surface	B1
9(b)(i)	threshold (frequency)	B1
9(b)(ii)	- photons are (discrete) packets of energy - energy of photons depends on frequency (of EM radiation) - electrons can only absorb a single photon (of energy) <i>Any two points, 1 mark each</i>	B2
	emission only possible if photon energy is at least the work function	B1
9(b)(iii)	work function = $hf_0 = 6.63 \times 10^{-34} \times 6.93 \times 10^{14}$	C1
	$= 4.59 \times 10^{-19} \text{ (J)}$ $= 4.59 \times 10^{-19} / 1.60 \times 10^{-19} \text{ (eV)}$ $= 2.87 \text{ eV}$	A1